Cambridge (CIE) A Level Maths: Pure 3

Your notes

Further Differentiation

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Your notes

Differentiating other functions

How do I differentiate common functions?

- These are the common results
  - $\frac{d}{dx}(x^n) = nx^{n-1}$
  - $\frac{d}{dx}(e^x) = e^x$
  - $\frac{d}{dx}(a^x) = a^x \ln a$ for $a > 0$
  - $\frac{d}{dx}(\ln x) = \frac{1}{x}$
  - $\frac{d}{dx}(\sin x) = \cos x$
  - $\frac{d}{dx}(\cos x) = -\sin x$
  - $\frac{d}{dx}(\tan x) = \sec^2 x$
  - $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
  - $\frac{d}{dx}(\sec x) = \sec x \tan x$
  - $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

How do I differentiate e^{kx}, a^x, ln(kx) and a^{kx}?

1. IF $y = e^{kx}$, WHERE k IS A REAL CONSTANT, THEN $\frac{dy}{dx} = ke^{kx}$
2. IF $y = \ln x$ THEN $\frac{dy}{dx} = \frac{1}{x}$

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3. IF $y = a^x$, WHERE a IS A REAL CONSTANT AND $a>0$, THEN $\frac{dy}{dx} = (\ln a)a^x$

4. IF $y = \ln kx$, WHERE k IS A REAL CONSTANT, THEN $\frac{dy}{dx} = \frac{1}{x}$

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i.e. IF $y = a^x$ THEN

WE CAN WRITE $a = e^{\ln a}$

REMEMBER THIS TRICK! e AND ln ARE INVERSES, SO THEY CANCEL EACH OTHER

THEN $a^x = (e^{\ln a})^x = e^{(\ln a)x}$

LAWS OF INDICES $(a^m)^n = a^{mn}$

SO $y = e^{(\ln a)x}$ AND

$\frac{dy}{dx} = (\ln a)e^{(\ln a)x} = (\ln a)a^x$

IF $y = e^{kx}$, $\frac{dy}{dx} = ke^{kx}$

$\ln a$ IS JUST A CONSTANT!

IF $y = \ln kx$ THEN

$y = \ln k + \ln x$

LAWS OF LOGARITHMS $\ln(ab) = \ln a + \ln b$

SO $\frac{dy}{dx} = \frac{1}{x}$

$\ln k$ IS A CONSTANT, SO ITS DERIVATIVE IS ZERO

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- And for a^{kx}:

5. IF $y = a^{kx}$, WHERE a AND k ARE REAL CONSTANT AND $a>0$,

THEN $\frac{dy}{dx} = k(\ln a)a^{kx}$

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- This last formula can be derived from Formula 3 by using the **chain rule**

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How do I differentiate reciprocal trigonometric functions?

- The formulae for the derivatives of the reciprocal trigonometric functions are:

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

- You can derive the derivatives for **sec**, **cosec**, and **cot** using the **chain rule** and the derivatives of the basic trigonometric functions

e.g. DIFFERENTIATE $y = \sec x$

$$y = \sec x = \frac{1}{\cos x} = (\cos x)^{-1}$$

USE DEFINITION OF sec

$$\frac{dy}{dx} = -(\cos x)^{-2}(-\sin x)$$

USE CHAIN RULE FOR $(f(x))^n$

$n(f(x))^{n-1}$

$f'(x)$

$$\frac{dy}{dx} = \frac{\sin x}{\cos^2 x} = \left(\frac{1}{\cos x}\right)\left(\frac{\sin x}{\cos x}\right)$$

$$\frac{dy}{dx} = \sec x \tan x$$

$\tan x = \frac{\sin x}{\cos x}$

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Examiner Tips and Tricks

All of the above are in the formula book, make sure you know how to find them. They will be particularly useful when working with identities and when integrating trigonometric functions.

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Worked Example

A curve has the equation $y = e^{-3x} + 3^{\frac{x}{2}} - \ln 5x$. Find the gradient of the curve at the point with x-coordinate 2.

$$\frac{dy}{dx} = -3e^{-3x} + \frac{1}{2}(\ln 3)3^{\frac{x}{2}} - \frac{1}{x}$$

FORMULA 1

FORMULA 5

FORMULA 4

WHEN $x = 2$,

$$\frac{dy}{dx} = -3e^{-6} + \frac{1}{2}(\ln 3)3^{1} - \frac{1}{2}$$

$$= \frac{1}{2}(3\ln 3 - 1) - 3e^{-6} \qquad (\approx 1.14)$$

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Product rule

What is the product rule?

- The product rule is a formula that allows you to differentiate a **product of two functions**
- If $y = u \times v$ where u and v are functions of x then the product rule is:

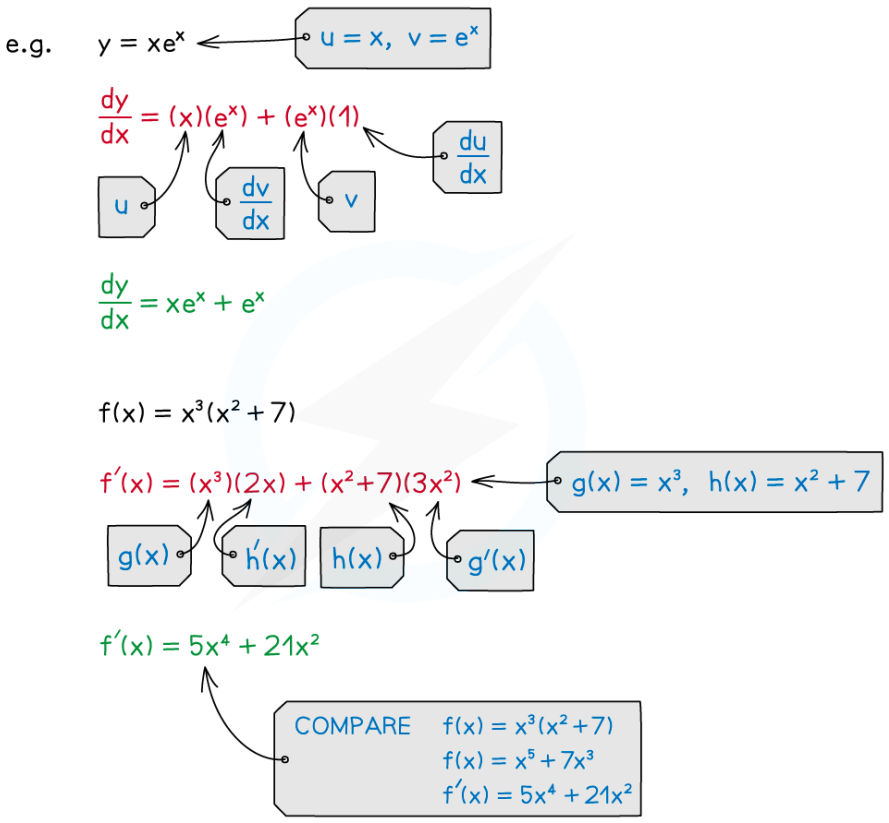
$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

- In **function notation**, if $f(x) = g(x) \times h(x)$ then the product rule can be written as:

$$f'(x) = g(x)h'(x) + h(x)g'(x)$$

- The easiest way to remember the product rule is, for $y = u \times v$ where u and v are functions of x:

$$y' = uv' + vu'$$



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Examiner Tips and Tricks

- The product rule formulae are in the formulae booklet – make sure you know how to find them.
- Don't confuse the **product** of two functions with a **composite function:**
  - The **product** of two functions is two functions **multiplied** together
  - A **composite** function is a **function of a function**

e.g. $f(x) = 3x^2$ $\quad g(x) = x - 3$

$f(x)g(x) = 3x^2(x-3) = 3x^3 - 9x^2$ PRODUCT

$fg(x) = f(g(x)) = 3(x-3)^2 = 3x^2 - 18x + 27$ COMPOSITE FUNCTION

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- To differentiate composite functions you need to use the **chain rule**

Worked Example

Your notes

Differentiate with respect to x:

a) $(5\sin x - 7\cos x)e^{3x+2}$

b) $5x\ln 5x$

a) $u = 5\sin x - 7\cos x$ $\qquad$ $v = e^{3x+2}$

$\frac{du}{dx} = 5\cos x + 7\sin x$ $\qquad$ $\frac{dv}{dx} = 3e^{3x+2}$

DON'T FORGET MINUS SIGN WHEN YOU DIFFERENTIATE cos

USE CHAIN RULE

$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

$$\frac{dy}{dx} = (5\sin x - 7\cos x)(3e^{3x+2}) + (e^{3x+2})(5\cos x + 7\sin x)$$

$$\frac{dy}{dx} = (22\sin x - 16\cos x)e^{3x+2}$$

b) $u = 5x$ $\qquad$ $v = \ln 5x$

$\frac{du}{dx} = 5$ $\qquad$ $\frac{dv}{dx} = \frac{1}{x}$

BE CAREFUL WITH THIS ONE! BECAUSE OF THE CHAIN RULE THE DERIVATIVE OF $\ln(ax)$ IS $\frac{1}{x}$ FOR **ANY** CONSTANT a

$$\frac{dy}{dx} = (5x)\left(\frac{1}{x}\right) + (\ln 5x)(5)$$

$$\frac{dy}{dx} = 5 + 5\ln 5x$$

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Quotient rule

What is the quotient rule?

- The quotient rule is a formula that allows you to differentiate a **quotient of two functions** (ie one function divided by another)
- If $y = \frac{u}{v}$ where u and v are functions of x then the quotient rule is:

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

- In **function notation**, if $f(x) = \frac{g(x)}{h(x)}$ then the quotient rule can be written as:

$$f'(x) = \frac{h(x)g'(x) - g(x)h'(x)}{(h(x))^2}$$

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e.g. $y = \dfrac{x^2+1}{x-3}$ ← $u = x^2+1, \quad v = x-3$

$$\frac{dy}{dx} = \frac{\overset{v}{(x-3)}\overset{\frac{du}{dx}}{(2x)} - \overset{u}{(x^2+1)}\overset{\frac{dv}{dx}}{(1)}}{\underset{v^2}{(x-3)^2}}$$

$$\frac{dy}{dx} = \frac{x^2-6x-1}{(x-3)^2}$$

$f(x) = \dfrac{\sin x}{\cos x}$ ← $g(x) = \sin x, \quad h(x) = \cos x$

$$f'(x) = \frac{\overset{h(x)}{(\cos x)}\overset{g'(x)}{(\cos x)} - \overset{g(x)}{(\sin x)}\overset{h'(x)}{(-\sin x)}}{\underset{h((x))^2}{(\cos x)^2}}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x}$$

$f'(x) = \sec^2 x$ ← $\dfrac{1}{\cos x} = \sec x$

$\dfrac{\sin x}{\cos x} = \tan x$ SO THIS IS THE DERIVATIVE OF $\tan x$

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Examiner Tips and Tricks

- The quotient rule formula is in the formulae booklet – you don't have to memorise it.
- Be careful using the formula – because of the minus sign in the numerator the order of the functions is important.
- Look out for functions of the form $\mathbf{f(x) = g(x)(h(x))^{-1}}$
  - You could differentiate that using a combination of the **chain rule** and the **product rule** (and it can be good practice for you to try it!)
  - But it can also be seen as just a quotient rule question in disguise

- $g(x)(h(x))^{-1} = \dfrac{g(x)}{h(x)}$

Your notes

Worked Example

Differentiate $f(x) = (\cos 2x)\left(x^2 + 2x + 1\right)^{-1}$ with respect to x.

$$f(x) = \frac{\cos 2x}{x^2 + 2x + 1}$$

REWRITE AS $f(x) = \dfrac{g(x)}{h(x)}$

$g(x) = \cos 2x \qquad h(x) = x^2 + 2x + 1$

$g'(x) = -2\sin 2x \qquad h'(x) = 2x + 2$

CHAIN RULE – AND REMEMBER MINUS SIGN WHEN DIFFERENTIATING cos

$$f'(x) = \frac{(x^2 + 2x + 1)(-2\sin 2x) - (\cos 2x)(2x + 2)}{(x^2 + 2x + 1)^2}$$

$f'(x) = \dfrac{h(x)g'(x) - g(x)h'(x)}{(h(x))^2}$

$$f'(x) = \frac{-2\sin 2x(x^2 + 2x + 1) - 2\cos 2x(x + 1)}{(x^2 + 2x + 1)^2}$$

$$= \frac{-2\sin 2x(x + 1)^2 - 2\cos 2x(x + 1)}{(x + 1)^4}$$

$x^2 + 2x + 1 = (x + 1)^2$

$$f'(x) = \frac{-2(\cos 2x + (x + 1)\sin 2x)}{(x + 1)^3}$$

SIMPLIFY IT

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